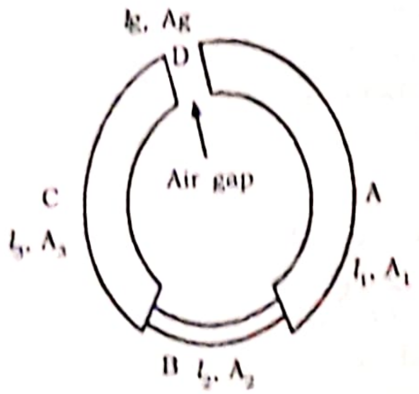
2.18 Composite Magnetic Circuit



In several cases, the magnetic circuit consists of several regions of different materials and different dimensions. The figure shows four regions A, B, C, D of lengths l_1, l_2, l_3 and l_g respectively. The area of cross-sections are A_1, A_2, A_3 and A respectively. Each of these regions has different reluctance. Then the total reluctance of the circuit is the sun of individual reluctances.

$$\text{Total reluctance} = \frac{l_1}{\mu_1 A_1} + \frac{l_2}{\mu_2 A_2} + \frac{l_3}{\mu_3 A_3} + \frac{l_g}{\mu_0 A_g}$$

where μ_1, μ_2, μ_3 and μ_0 are the permeability of A, B, C and air respectively.

$$\text{Flux } (\phi) = \frac{\text{MMF}}{\text{Total reluctance}}$$

Example 1

An iron ring having cross-sectional area of 400m m² and mean circumference of 500mm carries a coil of 250 turns wound uniformly around it. Calculate (a) Reluctance of the ring. (b) Current required to produce a flux of 1000 μ wb in the ring. Relative permeability of iron is 400.

Solution:

$$l = 500\text{mm} = 500 \times 10^{-3}\text{m}, \ N = 250$$

$$A = 400\text{mm}^2 = 400 \times 10^{-6}\ \text{m}^2, \ \mu_r = 400$$

$$\text{Reluctance (S)} = \frac{l}{\mu_0 \mu_r A} = \frac{500 \times 10^{-3}}{4\pi \times 10^{-7} \times 400 \times 400 \times 10^{-6}} = 2487 \times 10^3\ \text{AT/wb}$$

$$\text{MMF} = \text{Flux} \times \text{Reluctance}$$

$$\text{NI} = \phi S$$

$$\text{Hence} \quad I = \frac{\phi S}{N} = \frac{1000\times10^{-6}\times2487\times10^{-3}}{250} = \mathbf{9.948\times10^{-6}\ Amp}$$

Example 2

A mild steel ring has a mean circumference of 500mm and a uniform cross sectional area of 300mm². Calculate the mmf required to produce a flux of 500μwb? Assume $\mu_r = 1200$.

Solution:

$$l = 500\text{mm} = 500\times10^{-3}\text{m}$$

$$A = 300\text{mm}^2 = 300\times10^{-6}\text{m}^2$$

$$\phi = 500\ \mu\text{wb} = 500\times10^{-6}\text{wb}$$

$$\mu_r = 1200$$

$$\text{Reluctance (S)} = \frac{l}{\mu_0\mu_r A} = \frac{500\times10^{-3}}{4\pi\times10^{-7}\times1200\times300\times10^{-6}}$$

$$S = 1105242.\text{AT/wb}$$

$$\text{MMF} = \text{Flux}\times\text{Reluctance} = \phi\times S = 500\times10^{-6}\times1105242.6 = 552.62\ \text{AT}$$

Example 3

Six ampere current is flowing in a solenoid wound with 1200 turns of wire. If the length of the solenoid is 160cm. Calculate field strength of the solenoid?

Solution: $I = 6\text{ A},\ N = 1200,\ l = 160\text{cm} = 1.6\text{m}$

$$\text{Field Strength (H)} = \frac{NI}{l} = \frac{1200\times6}{1.6} = 4500\ \text{AT/m}$$

Example 4

An iron ring of mean length 60cm has an air gap of 2mm and a winding of 300 turns. If the relative permeability of iron used in the ring is 400 when a current of 1.5 A flows through it, find the flux density?

Solution:

Left the flux density be B wb/m²

MMF required for the iron ring is given by

$$\text{MMF}_1 = H_1 l_1 = \frac{B l_1}{\mu_0\mu_r} = \frac{B\times0.6}{\mu_0\times400}$$

MMF required for the air gap is given by,

$$MMF_2 = H_2 l_2 = \frac{Bl_2}{\mu_0\mu_r} = \frac{B\times 2\times 10^{-3}}{\mu_0\times 1} = \frac{B\times 2\times 10^{-3}}{\mu_0}$$

$$\text{Total MMF} = MMF_1 + MMF_2 = \frac{B\times 0.6}{\mu_0\times 400} + \frac{B\times 2\times 10^{-3}}{\mu_0} = \frac{B}{\mu_0}\left[\frac{0.6}{400} + 2\times 10^{-3}\right]$$

$$\text{Total MMF} = \frac{B}{\mu_0}\left[\frac{0.6}{400} + 2\times 10^{-3}\right] \quad \text{...........(1)}$$

$$\text{AT provided by current} = 1.5 \times 300 \quad \text{...........(2)}$$

Equating (1) and (2) we get

$$1.5\times 300 = \frac{B}{\mu_0}\left[\frac{0.6}{400} + 2\times 10^{-3}\right]$$

Hence, $\mathbf{B} = \mathbf{0.1615\ wb/m^2}$

Example 5

A circular iron ring has circular cross sectional area of 12 cm² and length cm in iron. An air gap of 1 mm is made by a sawcut. Find the ampere-tur needed to produce a flux of 24.84 μ wb. Relative permeability of iron is 800. Negle leakage and fringing?

Solution:

$$\phi = 24.84\ \mu wb = 24.84\times 10^{-6}\ wb$$

$$A = 12\ cm^2 = 12\times 10^{-4}\ m^2$$

$$B = \frac{\phi}{A} = \frac{24.84\times 10^{-6}}{12\times 10^{-4}} = 0.0207\ wb/m^2$$

(a) For air gap

$$H_a = \frac{B}{\mu_0} = \frac{0.0207}{4\pi\times 10^{-7}}$$

$$H_a = 16472.5\ AT/m$$

$$MMF_a = H_a \cdot l_a = 16472.5\times 1\times 10^{-3} = \mathbf{16.4725\ AT}$$

(b) Iron ring

$$H_i = \frac{B}{\mu_0\cdot\mu_r} = \frac{0.0207}{4\pi\times 10^{-7}\times 800}$$

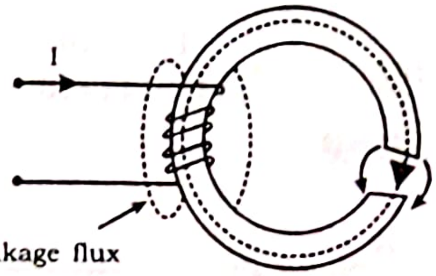
$$H_i = 20.59\ AT/m$$

$$MMF_i = H_i l_i = 20.59\times 0.15 = \mathbf{3.088\ AT}$$

Total ampere turns needed = $MMF_a + MMF_i$ = 16.4725 + 3.088 = **19.56 AT**

2.19 Leakage and Fringing in Magnetic circuit

The flux which does not follow the desired path in a magnetic circuit is called Leakage flux.



Consider an iron ring with air gap as shown in figure. Let a coil be wound on a portion of the ring. Let a current I flows through the coil. The complete flux produced in the ring by the current does not flow through the core. A small quantity of flux leaks through the air surrounding the iron ring. This flux is called Leakage flux. This leakage flux cannot be utilized for any purpose.

Total flux set up = Useful flux + Leakage flux

Leakage factor

It is defined as the ratio of total flux to the useful flux.

$$\text{Leakage factor } \lambda = \frac{\text{Total flux}}{\text{Useful flux}}$$

Fringing flux

Out of the total flux in air gap, a part of the flux diverges outside the main air gap. As a result, effective cross-sectional area of the air gap increases and the flux density in the air gap decreases. The diverging flux is called the **Fringing flux.**

Example 1

A circular iron ring having cross-sectional area of 20 cm^2 and length 30 cm in iron has an air gap of 2 mm made by a saw cut. Relative permeability of iron is 900. The ring is wound with a coil of 2500 turns and the current in the coil is 3A. Determine the air gap flux. Given that the leakage coefficient is 1.1.

Solution

Let the flux density through the iron be B. Then the flux density through the air is B/1.1

(a) Air gap

$$H_a = \frac{B}{\mu_0} = \frac{B/1.1}{4\pi \times 10^{-7}} \text{ AT/m}$$

$$l_a = 2 \times 10^{-3} \text{ m}$$

$$MMF_a = H_a l_a = \frac{B \times 2 \times 10^{-3}}{1.1 \times 4\pi \times 10^{-7}}$$

$$MMF_a = 1446 \text{ B}$$

(b) Iron ring

$$H_i = \frac{B}{\mu_0 \mu_r} = \frac{B}{4\pi \times 10^{-7} \times 900}$$

$$l_i = 0.3 \text{ m}$$

$$MMF_i = H_i l_i = \frac{B \times 0.3}{4\pi \times 10^{-7} \times 900}$$

$$MMF_i = 265 \text{ B}$$

Total MMF required $= MMF_a + MMF_i$

$= 1446 \text{ B} + 265 \text{ B} = 1711 \text{ B}$

But AT applied $= NI = 2500 \times 3 = 7500$

Hence $1711 \text{ B} = 7500$

$$B = \frac{7500}{1711} = 4.38 \text{ wb/m}^2$$

Area of cross section, A $= 20 \text{ cm}^2 = 20 \times 10^{-4} \text{ m}^2$

Flux through the air gap $= \frac{BA}{1.1} = \frac{4.38 \times 20 \times 10^{-4}}{1.1} = \mathbf{7.96 \times 10^{-3}}$ **wb**

2.20 Force Experienced by Current Carrying Conductor in a Magnetic Field

When a conductor of length 'l' metre is placed at right angles to a magnet[ic] field, it experiences a force F given by,

$$F = BIl \text{ Newton}$$

Where I = Current through the conductor in ampere

B = Flux density in Wb/m^2.

If the conductor is placed at an angle 'θ' to the direction of the magnetic field, the force is given by

$$F = BIl \sin \theta$$

Example 1

A straight conductor 2m long carries 30 A current and lies perpendicular to a uniform field of 0.5 wb/m^2. Find the force on the current carrying conductor?

Solution:

$$l = 2m, I = 30A, B = 0.5wb/m^2$$

$$F = BIl = 0.5 \times 30 \times 2 = \mathbf{30N}$$

Example 2

A straight conductor 1.5m long carries a current of 80A and lies at right angles to a uniform field of flux density 2 wb/m^2. Find the force on the conductor when (i) It lies in the given position (ii) It lies a position such that it is inclined at an angle of 30^0 to the direction of the filed?

(i) $F = BIl = 2 \times 80 \times 1.5 = 240N$

(ii) $F = BIl \sin\theta = 2 \times 80 \times 1.5 \times \sin 30^0 = \mathbf{120N}$

2.21 ENERGY STORED IN MAGNETIC FIELD

Consider a coil having a constant inductance of L henry, in which the current grows at a uniform rate from zero to I amperes in 't' seconds. The induced emf in the coil is given by,

$$e = -L \times \frac{(I-0)}{t} = -\frac{LI}{t} \text{ ....(1)}$$

The component of applied voltage required to neutralise this induced emf will be equal to L I/t.If we assume that the current increases by di in dt seconds, then Eq.(1) becomes

$$e = -L\frac{di}{dt}$$

The applied voltage must balance the voltage dropacross resister R and neutralize the above indused emf, thus,

$$V = iR + L\frac{di}{dt}$$

Multiplying throughout by i.dt

$$V\,i\,dt = i^2R\,dt + L\,i\,di$$

where,

V i dt, is the energy supplied by the source in time dt.

i^2 R dt, the energy dissipated in the form of heat.

Li di, the energy absorbed by the inductance of the coil in building up the magnetic field.

Thus, energy absorbed by the magnetic field during the time dt second

$$= Li\, di \text{ Joules}$$

Hence, total energy absorbed by the magnetic field when the current increases from zero to I amperes

$$= \int_0^I Li \cdot di = L\int_0^I i\, di$$

$$\text{Energy stored} = \frac{1}{2}LI^2$$

PROBLEMS

1. Find the inductance of an air - cored solenoid having a diameter of 6 cm. And a length of 40 cm, wound with 3000 turns.

Ans. Length of air-cored solenoid $= 40 \times 10^{-2}$ m

Diameter of solenoid $= 6 \times 10^{-2}$ m

No. of turns $= 3000$

Then,

Cross-sectional area, $A = \pi r^2$

$$= \pi \times \left(\frac{6 \times 10^{-2}}{2}\right)^2 = 2.827 \times 10^{-3}\ m^2$$

Slef inductance of solenoid 'L' $= N^2/S$

where 'N' is the number of turns and 'S' is the reluctance

$$S = \frac{l}{\mu_0 \mu_r \times A} \qquad \mu_r = 1$$

$$\text{Inductance } L = \frac{3000^2 \times 4\pi \times 10^{-7} \times 1 \times 2.827 \times 10^{-3}}{40 \times 10^{-2}} = \mathbf{0.0794\ H}$$

2. 12. A mild steel ring of 30 cm mean circumference has a cross sectional area of 6 cm^2 and has a winding of 500 turns on it. The ring is cut through at a point 50 as to provide an air gap of 1 mm in the magnetic circuit. It is found that a current of 4A in the winding produces a flues density of 1 T in the air gap. Find

(i) The relative permeability of the mild steel and

(ii) Inductance of the winding.

Ans. $l = 30 \times 10^{-2}$ m, $A = 6 \times 10^{-4}\ m^2$

$N = 500$, $l_g = 1 \times 10^{-3}$ m, $I = 4$ A

Air gap reluctance $S_g = \dfrac{l_g}{\mu_0\mu_r \times A}$

$$= \frac{1\times10^{-3}}{4\pi\times10^{-7}\times1\times6\times10^{-4}} = 1326291.19 \text{ AT/Wb}$$

Core flux $\phi = B \times A = 1 \times 6 \times 10^{-4} = 6 \times 10^{-4}$ Wb

Total reluctance $S = \dfrac{\text{mmf}}{\phi} = \dfrac{500\times4}{6\times10^{-4}} = 3333333.3$ AT/Wb

Reluctance of iron core, $S_i = S - S_g = 2007042.11$ AT/Wb

$$2007042.11 = \frac{30\times10^{-2} - 1\times10^{-3}}{4\pi\times10^{-7}\times\mu_r\times6\times10^{-4}}$$

$\mu_r = 197.58$

Inductance of the winding $= \dfrac{N^2}{S} = 0.075$ H

3. An iron ring of mean diameter 100 cm and a cross sectional area of 6 cm^2 is wound with 200 turns of wire. Calculate the current required to produce a flux of 0.6 mWb in the ring, if the relative permeability of iron is 2000. If now a radial cut of 2 mm is made in the iron ring, find the new value of current required to produce the same flux in the air gap. Neglect fringing and leakage flux.

Ans. Mean diamter of the ring (D) = 100 cm $= 100 \times 10^{-2}$ m

Length (l) $= \pi D = \pi \times 100 \times 10^{-2}$ m

Cross sectional area (A) = 6 cm^2 $= 6 \times 10^{-4}$ m^2

$N = 200$, $\mu_r = 2000$, $\phi = 0.6$ m Wb $= 0.6 \times 10^{-3}$ Wb

$$\text{Reluctance (s)} = \frac{l}{\mu_0\mu_r A} = \frac{\pi\times100\times10^{-2}}{4\pi\times10^{-7}\times2000\times6\times10^{-4}} = 2083.3 \times 10^3 \text{ AT/Wb}$$

MMF = Flux × Reluctance

$NI = \phi \times S$

$$\text{Hence } I = \frac{\phi\times S}{N} = \frac{0.6\times10^{-3}\times2083.3\times10^3}{200} = 6.249 \text{ A}$$

If now a radial cut of 2 mm is made in the iron ring. Find the new value of current required to produce the same flux in the air gap?

Mean length of the iron ring (l_i) $= \pi \times 100 \times 10^{-2} - 2 \times 10^{-3}$

$= 3.139$ m

Length of the air gap (l_a) $= 0.002$ m

$$\text{Reluctance of iron ring } (S_i) = \frac{l_i}{\mu_0\mu_r A} = \frac{3.139}{4\pi\times10^{-7}\times2000\times6\times10^{-4}}$$

$= 2081.614 \times 10^3$ AT/Wb

$$\text{Reluctance of air gap } (S_a) = \frac{l_a}{\mu_0\mu_r A} = \frac{0.002}{4\pi\times10^{-7}\times1\times6\times10^{-4}}$$

$= 2652.58 \times 10^3$ AT/Wb

Total Reluctance (S) $= S_i + S_a$

$= 2081.614 \times 10^3 + 2652.58 \times 10^3$

$= 4734.19 \times 10^3$ AT/Wb

MMF = Flux × Reluctance

$NI = \phi \times S$

Hence $I = \frac{\phi \times S}{N} = \frac{0.6 \times 10^{-3} \times 4734.19 \times 10^3}{200}$

$= 14.2$ A

4. A flux of 0.5 mWb produced by a coil of 900 turns wound on a ring with a current of 3A in it. Calculate (i) the inductance of the coil (ii) the emf induced in the coil, when a current of 5A is switched off, assuming the current to fall to zero 1 millisecond.

Ans. $\phi = 0.5 \times 10^{-3}$ Wb, N = 900, I = 3A

Inductance, $L = \frac{N\phi}{I}$

$= \frac{900 \times 0.5 \times 10^{-3}}{1} = 0.15$ H

If a current of 5 A changes to zero in 1 millisecond,

emf induced, $e = L \times \frac{di}{dt} = 0.15 \times \frac{(5-0)}{1 \times 10^{-3}} = 750$ V

5. A magnetic core in the form of a closed ring has a mean length of 20 cm and a cross section of 1 cm². The relative permeability of iron is 2400. What direct current will be needed in a coil of 2000 turns uniformly wound round the ring to create a flux of 0.2 mwb in the iron.

Ans. Mean length $= 20 \times 10^{-2}$ m

Area of cross section $= 1 \times 10^{-4}$ m²

$\mu_r = 2400$

$N = 2000$

$\phi = 0.2$ mWb

Reluctance 'S' of the core $= \frac{l}{\mu_0 \mu_r \times A}$

Inductance of the core $= N^2/S$

$= \frac{N^2 \times \mu_0 \mu_r A}{l}$

$= \frac{2000^2 \times \mu_0 \times 2400 \times 1 \times 10^{-4}}{20 \times 10^{-2}}$

$= 6.03$ H

Then, $I = \frac{N\phi}{L}$

$$= \frac{2000 \times 0.2 \times 10^{-3}}{6.03}$$

$$= 0.066 \text{ A}$$

PRACTICE PROBLEMS

1. A cast steel electromagnet has an air gap length of 3 mm and an iron path of length 40 cm. Find the number of ampere turns necessary to produce a flux density of 0.7 Wb/m^2 in the gap. Neglect leakage and fringing. [*Ans.* 1935 AT]

2. An iron ring has a cross-sectional area of 8×10^{-4} m^2 and a mean diameter of 38.2 cm. It is uniformly wound with a coil of 480 turns. When the coil is carrying a current of 2A, the flux set up in the ring is found to be 0.008 Wb. Find out the relative permeability of iron at this flux density. [*Ans.* 1250]

3. The airgap in a magnetic circuit is 1.5 mm long and 2500 mm^2 in cross section. Calculate (a) the reluctance of the air gap and (b) the mmf to send a flux of 800 μWb across the air gap.

   [*Ans.* (a) 0.477×10^6 AT/Wb, (b) 382 AT]

4. A solenoid 1 m in length and 10 cm in diameter has 5000 turns. Calculate the (a) the approximate inductance (b) the energy stored in the magnetic field when a current of 2 A flows in the solenoid. [*Ans.* (a) 0.246 H, (b) 0.492 J]

5. A current of 2A is flowing through each of the conductors in a coil containing 15 such conductors. If a point pole of unit strength is placed at a perpendicular distance of 10 cm from the coil, determine the field intensity at that point.

   [*Ans.* 47.74 AT/m]

6. A coil of 100 turns is wound uniformly over an isolator ring with a mean circumference 2 m and a uniform sectional area of 0.025 cm^2. If the coil carries current of 2A, calculate (a) the mmf of the circuit (b) magnetic field intensity (c)flux density and (d) total flux.

   [*Ans.* (a) 200 AT (b) 100 AT/m (c) 0.12565 m Wb/m^2 (d) 3.14×10^{-9} Wb]

7. An iron ring of mean circumference 50 cm has an air gap of 1 mm. It is uniformly wound with a coil having 200 turns. If the relative permeability of iron is 300, when a current of 1A flows through the coil, calculate the flux density.

   [*Ans.* 94.2 mWb/m^2]